

Research Interests

Galaxy formation and evolution; galaxy-halo connection; baryon census; multi-phase gas; multi-wavelength photometry; HI and X-ray stacking; large extragalactic surveys (GAMA, ASKAP, eROSITA, 4MOST/WAVES); feedback and gas accretion in galaxies; AGN-galaxy coevolution; cosmological simulations.

Education

- 2022–2026 **PhD in Astrophysics**, *International Centre for Radio Astronomy Research (ICRAR)*, University of Western Australia, Perth, Australia
(Passed) Supervisors: Prof. Simon Driver, Dr. Martin Meyer.
Thesis: *Hot Gas, Cold Gas, and Stars: Mapping the Halo Baryon Budget in the Local Universe.*
- 2016–2021 **Integrated Master of Science (Physics)**, *National Institute of Science Education and Research (NISER)*, Bhubaneswar, India, HBNI
Thesis: *Detection of WHIM between cluster pairs Using SZ Effect.*
DST-INSPIRE Fellow (Govt. of India, 2016–2021).

Academic Honours

- 2025 **Best Talk (2nd Place)**, Astronomical Society of Australia (ASA) Annual Scientific Meeting, Adelaide.
- 2022–2025 **PhD Scholarship**, The University of Western Australia
- 2016–2021 **DST-INSPIRE Fellowship**, Department of Science and Technology, Government of India.
- 2016–2017 **Academic Excellence Award**, NISER Bhubaneswar.

Surveys, Observing and Computing Experience

- Survey Membership GAMA, DEVILS, 4MOST/WAVES, 4MOST/4HS, ASKAP/DINGO, ASKAP/WALLABY, ASKAP/EMU, EMPOWER
- Observing Proposals Co-I, EMPOWER Survey (PI: Paola Popesso), VLT/KMOS, 150 hrs.
- Observing Nights ESO VISTA Telescope (4MOST, 10 nights at Paranal Chile, May 2026); Blanco Telescope (DECam, 2024, 2 nights, remote); GMRT (2017, 1 night).
- Computing Skills R (advanced), Python (intermediate), bash scripting.
- HPC Time Co-I (PI: Sabine Bellstedt), Pawsey Merit Allocation Scheme 2024–2025, 1M CPU hours, Pawsey Supercomputer.

Talks and Seminars

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| UTFSM Valparaiso, Chile | <i>Seminar, 2026</i> |
| UDP Santiago, Chile | <i>Seminar, 2026</i> |
| ESO Santiago, Chile | <i>Seminar, 2026</i> |
| Macquarie University, Sydney (ASA ECR Talk) | <i>Seminar, 2026</i> |
| Hamburg Observatory, Germany | <i>Seminar, 2025</i> |
| Argelander Institute for Astronomy, Germany | <i>Seminar, 2025</i> |

NRC Herzberg Astronomy and Astrophysics Research Centre, Canada	<i>Seminar, 2025</i>
University of Victoria, Canada	<i>Seminar, 2025</i>
Canadian Institute for Theoretical Physics, Canada	<i>Seminar, 2025</i>
Multiphase, Multi-temperature and Complex, Olbia, Italy	<i>Contributed Talk, 2025</i>
Cosmic Ecosystems, Perimeter Institute, Canada	<i>Contributed Talk, 2025</i>
Astronomical Society of Australia (ASA) Annual Meeting, Adelaide	<i>Contributed Talk, 2025</i>
Galaxy Evolution Coffee, ESO Munich, Germany	<i>Talk, 2024</i>
MIAPbP Workshop: "Some Like it Hot", Munich, Germany	<i>Contributed Talk, 2024</i>
AAL/eROSITA Workshop (online)	<i>Contributed Talk, 2024</i>
A Journey through Galactic Environments, Porto Ercole, Italy	<i>Contributed Talk, 2023</i>
ASTRO3D Annual Conference, Fremantle, Australia	<i>Contributed Talk, 2023</i>
ACAMAR HI Conference, Perth, Australia	<i>Contributed Talk, 2023</i>
Astronomical Society of Australia (ASA) Annual Meeting, Sydney	<i>Contributed Talk, 2023</i>

Community and Outreach

2022–2024	Organiser, Journal Club
2022–2024	Volunteer, ICRAR night sky & public outreach events.
2022, 2024	Volunteer, ASTROFEST
2023	Night sky observation session at 2 Rocks primary school.
2016–2021	NISER Astronomy Club outreach events.

Publication List (ADS link)

First-Author Publications

2026 <i>Published</i>	<i>DINGO/GAMA/WAVES: HI–Halo Mass Relation.</i> Dev, A., Meyer, M. J., Driver, S. P., Rhee, J., Nulsen, P., Westmeier, T. et al. MNRAS ADS link
2024 <i>Published</i>	<i>The baryon census and the mass-density of stars, neutral gas, and hot gas as a function of halo mass.</i> Dev, A., Driver, S. P., Meyer, M. J., Robotham, A., Obreschkow, D., Popesso, P., Comparat, J. MNRAS, 535, 2357–2374. ADS link
2023 <i>Published</i>	<i>Galaxy And Mass Assembly (GAMA): The group HI mass as a function of halo mass.</i> Dev, A., Driver, S. P., Meyer, M. J., Roychowdhury, S., Rhee, J., Stevens, A. R. H., Lagos, C. D. P., et al. MNRAS, 523, 2693–2709. ADS link

Co-Author Publications

2025 <i>Published</i>	<i>The eROSITA view on the halo mass–temperature relation: From low-mass groups to massive clusters.</i> Toptun, V., Popesso, P., Marini, I., Dolag, K., Lamer, G., Yang, X., Li, Q., Csizi, B., Lovisari, L., Ettori, S., Biffi, V., Vladutescu-Zopp, S., Dev, A., et al. A&A, 700, A167. ADS link
2025 <i>Published</i>	<i>The impact of assembly history on the X-ray detectability of halos: From galaxy groups to galaxy clusters.</i> Marini, I., Popesso, P., Dolag, K., Biffi, V., Vladutescu-Zopp, S., Castro, T., Toptun, V., de Isídio, N., Dev, A., et al. A&A, 698, A191. ADS link

- 2024 *Detecting galaxy groups populating the local Universe in the eROSITA era.*
Published Marini, I., Popesso, P., Lamer, G., Dolag, K., Biffi, V., Vladutescu-Zopp, S., Dev, A., et al.
A&A, 689, A7. [ADS link](#)
- 2024 *Average X-ray properties of galaxy groups. From Milky Way-like halos to massive clusters.*
Submitted Popesso, P., Marini, I., Dolag, K., Lamer, G., Csizi, B., Biffi, V., Dev, A., et al.
arXiv:2411.17120. [ADS link](#)
- 2024 *The hot gas mass fraction in halos. From Milky Way-like groups to massive clusters.*
Submitted Popesso, P., Biviano, A., Marini, I., Dolag, K., Vladutescu-Zopp, S., Csizi, B., Biffi, V.,
Lamer, G., Dev, A., et al.
arXiv:2411.16555. [ADS link](#)
- 2024 *The perils of stacking optically selected groups in eROSITA data: The Magneticum perspec-*
Submitted *tive.*
Popesso, P., Marini, I., Dolag, K., Lamer, G., Csizi, B., Vladutescu-Zopp, S., Biffi, V., Dev,
A., et al.
arXiv:2411.16546. [ADS link](#)
- 2019 *Type IIP Supernova Progenitors and Their Explodability I: Convective Overshoot, Blue Loops*
Published *and Surface Composition.*
Wagle, G., Ray, A., Dev, A., et al.
ApJ, 886, 27. [ADS link](#)
- 2021 *VizieR Online Data Catalog: SNe IIP progenitors. I. LMC giant comparison sample (Wagle+,*
Published *2019).*
Wagle, G. A., Ray, A., Dev, A., Raghu, A.
VizieR On-line Data Catalog: J/ApJ/886/27. Originally published in: 2019ApJ...886...27W.
[ADS link](#)